



University of
Roehampton
London

MSc Project (CMP-L050-0)

MSc Computing
MSc Web Development
MSc Software Engineering
MSc Data Science
MSc Cyber Security

Workshop 1

Top 15%



Times
Higher
Education

Module Leaders



Research Interest:
Machine learning, AI,
computer vision, and
audio processing

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Research Interest:

Machine learning, AI,
computer vision, and data
mining, optimisation, fuzzy
logic.

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Agenda



- Module Introduction and Learning Outcomes
- Assessments
- What is Research, Research Process
- Types of research
- Bunch of Activities

Workshop Goals:



By the end of the workshop, you will have:

- A tentative research topic
- A clear problem statement
- Draft research questions
- An understanding of research types and methods aligned with their project

Module Introduction



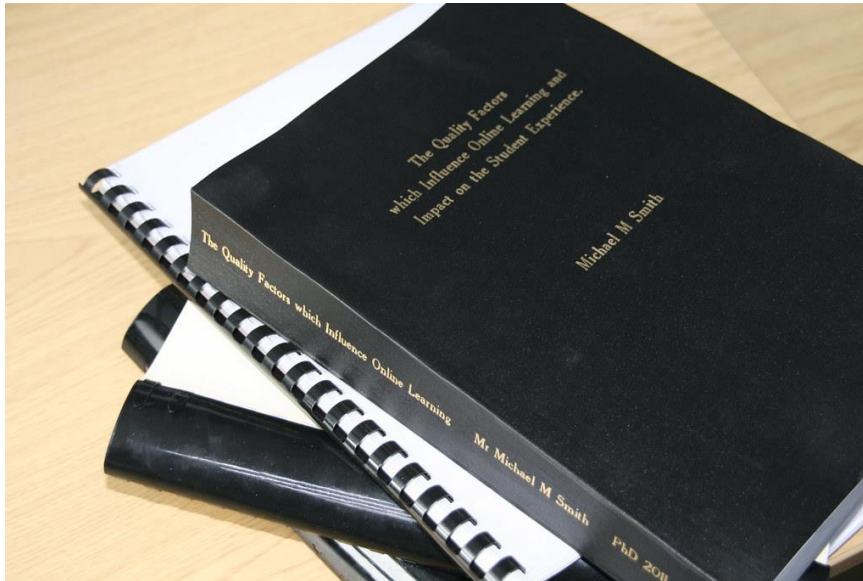
- A self-managed activity
- Identify a *research* project topic
- Define research aim(s) and objective(s)
- Develop and maintain project plan
- Project supervision
 - 12 Weeks of Supervision

Module Learning Outcomes



- **Self-manage** a significant **piece of individual work**
- **Synthesise** information, ideas, and practices.
- **Apply practical** and **analytical skills** in computer science
- **Produce** a project **deliverable** in a real-world context
- Critically **self-evaluate** the project process and deliverables.
- **Recognise** legal, social, ethical, and professional **issues** relevant to a **project**.
- **Summarise** the entire project **deliverable** and **process, results**.

Assessment



DISSERTATION (100%)
Final Report
Due Date: **20 April 2026 6:00 PM**

What is Research?



“Knowledge is of two kinds:
we **know a subject ourselves**, or
we **know where** we can find information about it.”

Samuel Johnson 1709-1784

What is Research?



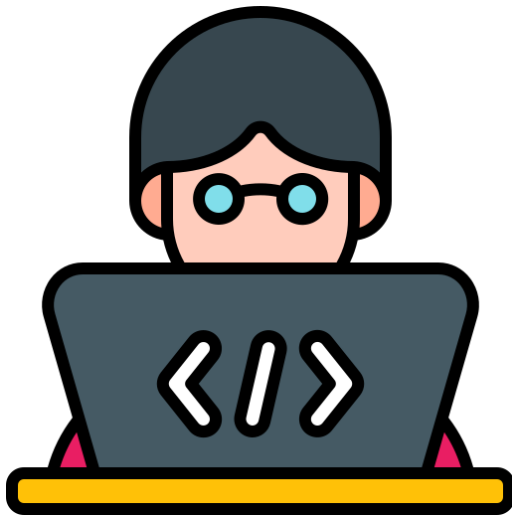
- A **systematic investigation or inquiry** aimed at **discovering, interpreting, and expanding knowledge** in a particular field of study or **solving a specific problem**.
- It involves **gathering, analysing, and evaluating information or evidence** through various methodologies, such as **experimentation, observation, data analysis, and theoretical reasoning**.
- The **ultimate goal** of research is to **contribute new insights, theories, solutions, or advancements to existing knowledge or practices** within academia, industry, or **society as a whole**.

What is Research?



- Usually conducted under the guidance of **research supervisors**.
- **MSc Project module** provides you with **opportunity** to acquire an **important skill** of **identifying a problem** and **solving it systematically**.

Activity 1

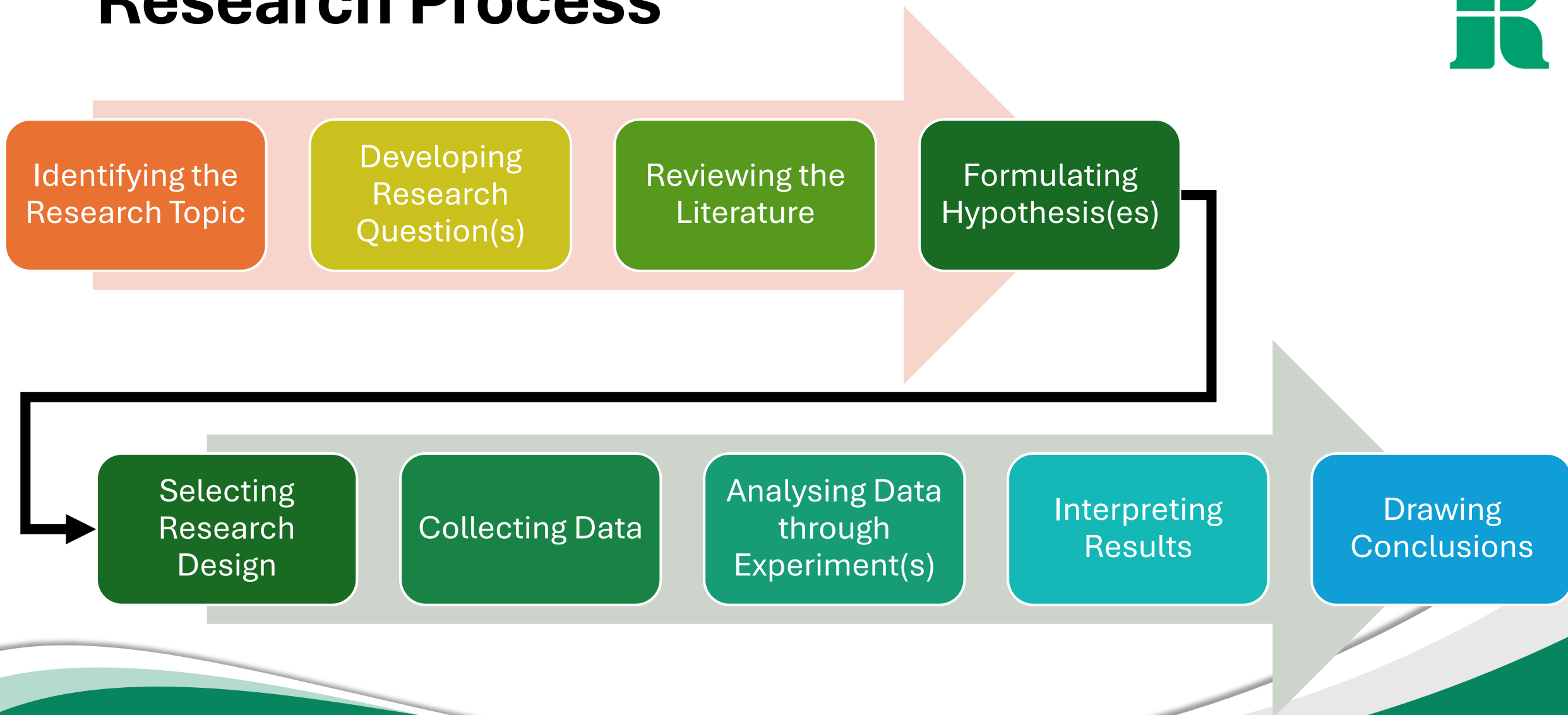


Mentimeter

For each of the project topics, identify if it is a research project.

Select True if so; otherwise, select False.

Research Process



Activity 2



Group Activity:

- Divide yourself into groups.
 - Select one of the following research topics.
 - Develop a flow diagram or bullet list showing how the project would move from idea to results.
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- a) *Explainable deep behavioural sequence clustering for transaction fraud detection*
 - b) *A new hybrid approach for intrusion detection using agentic AI*
 - c) *Automating and optimizing software testing using artificial intelligence techniques*
 - d) *Comparative analysis of energy-efficient scheduling algorithms for cloud computing*

Research Question



- **A question** that your research project, through its **thesis statement, aims to answer.**
- It is a **specific inquiry** that **guides** a research study or investigation.
- Research question(s) are formulated **to address a particular problem, issue, or gap in knowledge** within a specific field of study.
- Your research question(s) should be **clear, concise, and answerable** through **empirical research methods.**

Research Question



- Few examples:

Dissertation Title

An Innovative Approach to Diabetes Detection: Leveraging Machine Learning Techniques

1.5 Research Questions

To guide the study and ensure an absolute assessment of the problem domain, the following research questions (RQs) have been designed and are tackled in the course of this study:

RQ 1. How accurately does the proposed system predict diabetes using machine learning as compared to traditional methods?

Rationale: The aim is to evaluate the performance with metrics like Accuracy, F1-Score, Cohen Kappa, AUC-ROC, and Precision of the model.

RQ 2. How does integrating different datasets impact the accuracy of diabetes prediction?

Rationale: The objective is to understand how the combination of the different datasets in the model affects its prediction performance.

RQ 3. How can the system interpret predictions using explainable AI techniques?

Rationale: In answering this question, the project aims to evaluate how well users can understand the features the model considers important in its prediction.

RQ 4. How effective is the engagement of the conversational AI with the end user when communicating health-related issues concerning diabetes?

Rationale: The study aims to evaluate the performance of the chatbot with user satisfaction with information provided by the chatbot in response to respective queries.

RQ 5. How will user engagement and awareness of diabetes be influenced by real-time news and research updates?

Rationale: The study in this case aims to evaluate how updated information influences user behavior, understanding, and management of diabetes.

RQ 6. How well does the system design and development align with standard AI practice?

Rationale: The aim is to evaluate how well the project follows all ethical guidelines.

Research Question



- Few examples:

Dissertation Title

**AI-Powered Resume Analysis for
HR Recruitment**

1.6 Research Questions

1. How effective is the application of K-Means unsupervised machine learning for resume classification and subsequent job matching of candidate selection?
2. What are the system performance metrics and outcomes in terms of accuracy, efficiency and reliability of the system?
3. How can the AI-powered resume analysis system be refined and improved for scalability and further development and what are the recommendations for future directions in AI integration within HR recruitment practices?

Activity 3



From topic to research question:

• **Step 1: Brainstorm Topics (Individual)**

Write **3 topics** aligned with your specialization/interest.

Examples:

- Cyber Security: ransomware detection, zero-trust networks
- Data Science: customer churn prediction, sentiment analysis
- Software Engineering: software performance optimization, accessibility
- Computing: algorithms, AI models, system optimization

Step 2: Convert Topic into Questions (Pairs)

Examples:

- Topic: *Machine learning in cyber security*
- RQ: *How effective is a Random Forest model in detecting phishing websites compared to logistic regression?*

Step 3: Peer Feedback

Peers evaluate questions using a checklist:

- Is it specific?
- Can it be answered within MSc scope?
- Does it imply data or evaluation?

Problem Statement



- A problem statement is a **concise description of the issues or challenges** that a research **study aims to address**.
- It provides a **clear explanation** of the problem or gap in knowledge that the **research intends to investigate**.
- It helps to **establish the rationale** for the study and **guide the research objectives and methodology**.
- The purpose of a problem statement is to **define the problem** in a way that allows for **focused research, analysis, and problem-solving efforts**.
- It should highlight **problem, methodology** to solve the problem, and **research aim**; and must comply with **research questions**.

Problem Statement



- A well-written problem statement typically includes the following components:
 - Description of the **current situation** or **problem**.
 - Explanation of the **significance** or **impact**.
 - Clear specification of **scope** and **boundaries** (limitations or constraints of the problem).
 - **Measurable** or **observable** aspects.
 - Desired **outcome** or **solution**.

Activity 4



Problem Statement:

- Follow the given template, write a problem statement of about 200 words:
 - ✓ Context: What is the domain?
 - ✓ Problem: What exactly is not working well?
 - ✓ Evidence: What suggests this is a real issue?
 - ✓ Gap: What is missing in existing approaches?
 - ✓ Impact: Why does this problem matter?

Example (Cyber Security):

Despite the widespread use of email filters, phishing attacks continue to increase. Existing rule-based systems fail to detect sophisticated phishing emails, leading to financial and data losses. This highlights the need for more adaptive and intelligent detection techniques....



Empirical Research

vs.



Analytical Research



- Involves **collection and analysis of data** derived from **direct observation** or **experimentation**.
- Relies on **evidence** obtained through **observation** or **experience** and aims to **test hypotheses** or **answer research questions** through **systematic** data collection and analysis.
- Typically employs **quantitative** or **qualitative methods**, or a **combination** of both, to gather data.
- Objective is to **describe, explain, or predict phenomena** based on **observable evidence**.

- Involves the **critical analysis** and **interpretation** of **existing information, theories, or data** to gain insights.
- Relies on **deductive reasoning** and **logical interpretation** to **analyse existing information** or data sets.
- Involves **literature reviews, meta-analyses, theoretical frameworks, or mathematical models** to **evaluate** theories, concepts, or hypotheses.
- Objective is to **deepen understanding, generate new perspectives, or offer explanations** based on existing knowledge.



Qualitative Research

vs.



Quantitative Research



- Focuses on **exploring** and **understanding social phenomena, human experiences, and behaviours** through **non-numerical data**.
- Employs methods such as **interviews, focus groups, observations, and textual analysis** to collect rich, descriptive data.
- Objective is to **gain insights** into the **complexities and nuances of phenomena** by **examining the context, meanings, and social interactions** associated with them.
- Aims to **generate theories, hypotheses, or explanations** grounded in **real-world experiences**.

- Focuses on **measuring** and **analysing numerical data** to **quantify** relationships, patterns, and variables.
- Employs methods such as **surveys, experiments, observations, and statistical analyses** to collect **structured, numerical data**.
- Objective is to **describe, predict, or explain phenomena** by **quantifying variables** and **examining statistical relationships between them**.
- Aims to **generalize findings** to **larger populations** and **make objective conclusions** based on **empirical evidence**.

Activity 5



Mentimeter

Identify type of research

What's Next



- Read academic papers and journals
- Choose a project topic
- Complete the project outline
- Defining a research problem
- Research and project methods
- Collect and organise sources
- Referencing (Harvard SU)

Thanks!

Any Question...

